MSG No. 193.253 $P6_3/mcm$ [Type I, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-62m`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,3,5,7,8,9,14,16,18,22,23,24]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[2,4,6,10,11,12,13,15,17,19,20,21]`

Table 2: Wyckoff site: `2b`, site symmetry: `-3.m`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,3,5,10,11,12,13,15,17,22,23,24]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,4,6,7,8,9,14,16,18,19,20,21]`

Table 3: Wyckoff site: `4c`, site symmetry: `-6..`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,3,5,14,16,18]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[2,4,6,13,15,17]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[7,8,9,22,23,24]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[10,11,12,19,20,21]`

Table 4: Wyckoff site: `4d`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,3,5,10,11,12]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[2,4,6,7,8,9]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[13,15,17,22,23,24]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[14,16,18,19,20,21]`

Table 5: Wyckoff site: `4e`, site symmetry: `3.m`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,3,5,22,23,24]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,4,6,19,20,21]`
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[7,8,9,14,16,18]`
4	$[0,\ 0,\ -z]$	`[10,11,12,13,15,17]`

Table 6: Wyckoff site: `6f`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,11,13,23]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,9,14,21]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[3,12,15,24]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,7,16,19]`
5	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,10,17,22]`
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,8,18,20]`

Table 7: Wyckoff site: `6g`, site symmetry: `m2m`

No.	position	mapping
1	$\left[x, 0, \frac{1}{4}\right]$	`[1,7,16,23]`
2	$\left[x, x, \frac{3}{4}\right]$	`[2,10,17,21]`
3	$\left[0, x, \frac{1}{4}\right]$	`[3,8,18,24]`
4	$\left[-x, 0, \frac{3}{4}\right]$	`[4,11,13,19]`
5	$\left[-x, -x, \frac{1}{4}\right]$	`[5,9,14,22]`
6	$\left[0, -x, \frac{3}{4}\right]$	`[6,12,15,20]`

Table 8: Wyckoff site: `8h`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,3,5]`
2	$\left[\frac{2}{3}, \frac{1}{3}, z+\frac{1}{2}\right]$	`[2,4,6]`
3	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}-z\right]$	`[7,8,9]`
4	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[10,11,12]`
5	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[13,15,17]`
6	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}-z\right]$	`[14,16,18]`
7	$\left[\frac{1}{3}, \frac{2}{3}, z+\frac{1}{2}\right]$	`[19,20,21]`
8	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[22,23,24]`

Table 9: Wyckoff site: `12i`, site symmetry: `..2`

No.	position	mapping
1	$[x, 2x, 0]$	`[1,11]`
2	$\left[-x, x, \frac{1}{2}\right]$	`[2,9]`
3	$[-2x, -x, 0]$	`[3,12]`
4	$\left[-x, -2x, \frac{1}{2}\right]$	`[4,7]`
5	$[x, -x, 0]$	`[5,10]`
6	$\left[2x, x, \frac{1}{2}\right]$	`[6,8]`
7	$[-x, -2x, 0]$	`[13,23]`
8	$\left[x, -x, \frac{1}{2}\right]$	`[14,21]`
9	$[2x, x, 0]$	`[15,24]`
10	$\left[x, 2x, \frac{1}{2}\right]$	`[16,19]`
11	$[-x, x, 0]$	`[17,22]`
12	$\left[-2x, -x, \frac{1}{2}\right]$	`[18,20]`

Table 10: Wyckoff site: `12j`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,16]`
2	$\left[x-y,\ x,\ \frac{3}{4}\right]$	`[2,17]`
3	$\left[-y,\ x-y,\ \frac{1}{4}\right]$	`[3,18]`
4	$\left[-x,\ -y,\ \frac{3}{4}\right]$	`[4,13]`
5	$\left[-x+y,\ -x,\ \frac{1}{4}\right]$	`[5,14]`
6	$\left[y,\ -x+y,\ \frac{3}{4}\right]$	`[6,15]`
7	$\left[x-y,\ -y,\ \frac{1}{4}\right]$	`[7,23]`
8	$\left[y,\ x,\ \frac{1}{4}\right]$	`[8,24]`
9	$\left[-x,\ -x+y,\ \frac{1}{4}\right]$	`[9,22]`
10	$\left[x,\ x-y,\ \frac{3}{4}\right]$	`[10,21]`
11	$\left[-x+y,\ y,\ \frac{3}{4}\right]$	`[11,19]`
12	$\left[-y,\ -x,\ \frac{3}{4}\right]$	`[12,20]`

Table 11: Wyckoff site: `12k`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,23]`
2	$\left[x,\ x,\ z+\frac{1}{2}\right]$	`[2,21]`
3	$[0,\ x,\ z]$	`[3,24]`
4	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	`[4,19]`
5	$[-x,\ -x,\ z]$	`[5,22]`
6	$\left[0,\ -x,\ z+\frac{1}{2}\right]$	`[6,20]`
7	$\left[x,\ 0,\ \frac{1}{2}-z\right]$	`[7,16]`
8	$\left[0,\ x,\ \frac{1}{2}-z\right]$	`[8,18]`
9	$\left[-x,\ -x,\ \frac{1}{2}-z\right]$	`[9,14]`
10	$[x,\ x,\ -z]$	`[10,17]`
11	$[-x,\ 0,\ -z]$	`[11,13]`
12	$[0,\ -x,\ -z]$	`[12,15]`

Table 12: Wyckoff site: `24l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	`[2]`
3	$[-y,\ x-y,\ z]$	`[3]`
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[4]`
5	$[-x+y,\ -x,\ z]$	`[5]`
6	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	`[6]`
7	$\left[x-y,\ -y,\ \frac{1}{2}-z\right]$	`[7]`
8	$\left[y,\ x,\ \frac{1}{2}-z\right]$	`[8]`
9	$\left[-x,\ -x+y,\ \frac{1}{2}-z\right]$	`[9]`
10	$[x,\ x-y,\ -z]$	`[10]`
11	$[-x+y,\ y,\ -z]$	`[11]`
12	$[-y,\ -x,\ -z]$	`[12]`

continued ...

Table 12

No.	position	mapping
13	$[-x,\ -y,\ -z]$	[13]
14	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[14]
15	$[y,\ -x+y,\ -z]$	[15]
16	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[16]
17	$[x-y,\ x,\ -z]$	[17]
18	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[18]
19	$\left[-x+y,\ y,\ z+\frac{1}{2}\right]$	[19]
20	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[20]
21	$\left[x,\ x-y,\ z+\frac{1}{2}\right]$	[21]
22	$[-x,\ -x+y,\ z]$	[22]
23	$[x-y,\ -y,\ z]$	[23]
24	$[y,\ x,\ z]$	[24]